MILITARY SPECIFICATION SHEET

TUBING, PLASTIC, FLEXIBLE, CONVOLUTED, POLYTETRAFLUOROETHYLENE, STANDARD CONVOLUTIONS

This specification has been approved by the Naval Air Systems Command, Department of the Navy

The complete requirements for procuring the tubing described herein shall consist of this document and the issue in effect of MIL-T-81914(AS).

REQUIREMENTS:

Convolution type: Helical (See 3.3)

Construction details. Construction details, including available sizes, dimensions and tolerances are located in Table I. Lengths shall be as specified by the procuring activity.

Continuous operating temperature: -67°C (-88°F) to +260°C (500°F)

Color. Unless otherwise specified, the supplied color shall be black.

Physical properties: General physical requirement values along with associated test conditions are located in Table II.

Part number: Consists of the basic number of this specification sheet and dash numbers as shown below:

M81914/1-0000

- Identifier (Table I)
- Class, when applicable (when no classes are specified, the numeric 0 shall be used)
- Type of convolution 1- helical, 2- annular
- Specification sheet

Standard convolution Polytetrafluoroethylene, standard wall, 0.500 inch is identified as: M81914/1-1106.

FSC 9330

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TABLE I

CONSTRUCTION DETAILS

Identifier	Max. Inside Dia.	Min. Inside Dia.	Max. Outer Dia.	Wall Thickness Max.	Convolutions Per Inch ± 1	Weight (Lbs) Per 100 feet Max.	Min. Bend Radius
**01	.188	.181	.320	.023	8	2.0	.500
**02	.281	.273	.414	.027	7 1/2	2.9	.750
**03	.312	.303	.450	.027	7 1/2	3.6	.875
**04	.375	.364	.530	.029	7	4.2	1.000
**05	.437	.425	.590	.029	7	4.9	1.250
**06	.500	.485	.660	.029	7	5.2	1.500
**07	.625	.608	.780	.035	7	6.9	1.750
**08	.750	.730	.975	.035	6	10.4	1.875
**09	.875	.850	1.100	.035	6	11.3	2.250
**10	1.000	.975	1.260	.035	4 1/2	12.6	2.500
**11	1.125	1.105	1.390	.035	4 1/2	13.8	2.750
**12	1.250	1.210	1.539	.035	4	15.5	3.000
**13	1.500	1.440	1.850	.040	4	21.7	3.750
**14	1.750	1.690	2.100	.045	4	25.3	4.250
**15	2.000	1.940	2.350	.045	4	29.0	4.750

** - The asterisks shall be replaced with convolution type and class designation.

Note: Unless otherwise specified all dimensions are in inches.

TABLE II

PHYSICAL PROPERTIES

PROPERTY	REQUIREMENT	TEST PARAGRAPH
Construction details	In accordance with Table I	4.6.1
Stress in psi. at 10% strain	250 to 900	4.6.2
Specific gravity, max.	2.20	4.6.3
Crush resistance, pounds, horizontal min.	10	4.6.4

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TABLE II (Continued)

PROPERTY	REQUIREMENT	TEST PARAGRAPH
Low temperature flexibility	No cracking	4.6.5; 20,000 cycles
Heat shock % longitudinal chge, max.	No dripping, flowing or cracking 10	4.6.6; 232 ±2°C (450 ±4°F) for 4 hours
Heat aging Stress in psi @ 10% strain, % of initial value	80	4.6.7; 232 ±2°C (450 ±4°F) for 96 hours
Fluid resistance Stress in psi @ 10% strain, % of initial value	80	4.6.8
Flammability	Non-flammable 1/	4.6.9
Fungus resistance	Non-nutrient 1/	4.6.10
Dielectric breakdown	No breakdown, 12,000 volts	4.6.11

1/ See 6.4, Properties not tested for.

Standard procurement: Procured on coils. Unless otherwise specified, individual lengths shall be 18 feet.

Preparing Activity
Navy - AS
(Project No. 9330-N548-1)

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5 PROBLEM AREAS

a. Paragraph Number and Wording.

b. Recommended Wording

c Reason/Rationale for Recommendation·

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